

KOLYMSKOYE, Russian Federation

WMO: 251210

Lat: 68.729N

Lon: 158.704E

Elev: 12

StdP: 101.18

Time Zone: 11.00 (E11)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.2	-43.7	-47.7	0.0	-44.9	-46.2	0.0	-43.6	8.5	-21.9	7.8	-21.9	1.4	180	0.915

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	24.2	15.9	21.7	14.5	19.6	13.5	16.7	21.8	15.4	20.2	14.1	18.8	3.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	10.5	18.8	13.2	9.5	17.3	11.6	8.5	16.6	47.2	22.0	43.4	20.2	39.8	18.8	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.9	6.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-46.5	18.9	2.0	1.5	-48.0	19.9	-49.2	20.8	-50.3	21.6	-51.8	22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.1	-32.5	-31.4	-25.8	-13.3	-2.9	8.1	11.3	9.3	3.3	-8.4	-22.3	-30.2		
	DBStd	17.36	6.80	6.98	7.30	6.46	6.06	5.49	4.64	4.32	3.83	6.18	7.78	7.98		
	HDD10.0	7879	1316	1159	1110	698	400	102	37	67	206	571	968	1246		
	HDD18.3	10757	1574	1392	1368	948	658	308	224	282	451	829	1218	1504		
	CDD10.0	174	0	0	0	0	1	45	78	45	5	0	0	0		
	CDD18.3	10	0	0	0	0	0	2	7	2	0	0	0	0		
	CDH23.3	108	0	0	0	0	0	28	68	12	0	0	0	0		
	CDH26.7	17	0	0	0	0	0	3	13	1	0	0	0	0		
(14) Wind	WSAvg	3.3	2.8	2.9	3.0	3.2	3.7	3.9	3.5	3.6	3.6	3.2	3.2	3.4		
(15) Precipitation	PrecAvg	347	18	16	16	11	16	31	49	45	44	42	36	24		
	PrecMax	538	42	55	36	26	27	58	87	87	104	99	125	55		
	PrecMin	224	3	2	3	0	3	9	8	12	6	15	11	6		
	PrecStd	83	10	11	9	6	6	13	22	19	22	21	21	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.2	-9.0	-6.7	3.7	15.5	26.6	28.3	24.8	18.0	4.4	-2.0	-5.9		
		MCWB	-11.1	-10.0	-7.8	0.7	9.4	14.8	17.9	17.2	11.6	3.5	-2.4	-6.4		
	2%	DB	-16.5	-15.5	-9.5	2.1	11.1	23.4	24.9	21.9	14.3	1.9	-4.5	-12.3		
		MCWB	-16.8	-15.8	-10.4	-0.4	6.3	14.7	16.0	15.2	10.3	1.4	-5.0	-12.5		
	5%	DB	-19.7	-19.1	-12.0	-0.1	8.3	21.1	22.4	19.7	11.4	0.7	-7.9	-16.4		
		MCWB	-20.0	-19.3	-12.8	-1.8	4.3	13.0	15.1	14.2	7.8	0.4	-8.2	-16.7		
	10%	DB	-22.9	-21.3	-14.9	-3.1	5.9	18.6	19.7	17.1	9.0	-0.2	-11.3	-19.0		
		MCWB	-23.2	-21.6	-15.4	-4.6	3.0	11.8	14.1	13.2	6.5	-0.5	-11.5	-19.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.8	-9.8	-7.6	1.1	9.6	17.0	18.9	18.2	12.4	3.6	-2.3	-6.3		
		MCDB	-10.5	-8.9	-6.7	3.2	15.3	23.5	24.8	22.8	16.9	4.5	-2.1	-6.1		
	2%	WB	-16.7	-15.8	-10.3	-0.3	6.5	14.8	17.0	16.7	10.5	1.5	-4.9	-12.6		
		MCDB	-16.5	-15.6	-9.5	1.8	10.8	22.0	22.7	20.8	13.7	1.8	-4.5	-12.4		
	5%	WB	-19.8	-19.4	-12.7	-2.0	4.6	13.7	15.8	15.1	8.5	0.5	-8.2	-16.7		
		MCDB	-19.5	-19.1	-12.0	0.0	7.9	20.4	20.4	18.8	11.1	0.7	-7.9	-16.6		
	10%	WB	-23.1	-21.6	-15.4	-4.6	3.0	12.3	14.6	13.3	6.4	-0.5	-11.4	-19.3		
		MCDB	-22.9	-21.4	-14.9	-3.2	5.8	18.1	19.2	16.4	8.5	-0.3	-11.2	-19.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.0	8.9	10.3	7.7	9.1	7.9	8.0	5.5	4.7	6.6	6.4		
		MCDBR	9.3	9.4	11.2	11.5	10.9	14.0	12.7	13.4	10.0	4.1	8.2	10.3		
		MCWBR	9.2	9.2	10.3	9.6	6.6	6.9	5.5	6.8	6.0	3.8	8.1	9.8		
		MCDBR	9.4	9.1	10.8	11.3	10.7	13.6	11.8	11.4	9.2	3.9	8.3	9.9		
	5% WB	MCWBR	9.3	9.0	10.1	9.5	6.8	7.5	6.3	7.1	6.0	3.7	8.2	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.166	0.221	0.288	0.336	0.374	0.357	0.415	0.389	0.317	0.255	0.182	N/A		
	taud		1.920	2.063	2.072	2.019	2.186	2.430	2.255	2.296	2.461	2.144	1.922	N/A		
	Ebn at Noon		84	566	719	787	804	842	763	738	720	496	50	0		
	Edh at Noon		6	44	85	123	118	96	111	95	61	40	5	0		
(44) All-Sky Solar Radiation	RadAvg		0.02	0.47	2.18	4.58	5.85	5.73	4.87	3.40	1.76	0.57	0.08	0.00		
	RadStd		0.00	0.05	0.10	0.21	0.46	0.45	0.49	0.34	0.19	0.10	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+1.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page